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SAFETY MECHANISM FOR A DRUG DELIVERY DEVICE

Abstract

A safety mechanism for a drug delivery device for dispensing a liquid drug from a reservoir through a needle includes a housing defining a longitudinal axis, a needle cover movable along the longitudinal axis between a needle covering position and a retracted position, a movable lock element with a locking position to prevent the needle cover from being moved out of the needle covering position, and a trigger member movable along the longitudinal axis relative to the housing. The trigger member is operatively coupled to the lock element and a movement of the trigger member along the longitudinal axis causes the lock element to move out of the locking position allowing the needle cover to be moved out of the needle covering position.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a Continuation of U.S. patent application Ser. No. 19/033,106, filed on Jan. 21, 2025, which in turn is a Continuation of International Patent Application No. PCT/EP2023/068250, filed Jul. 3, 2023, which in turn claims priority to European Patent Application No. 22186390.5, filed Jul. 22, 2022, each of which is incorporated by reference herein, in the entirety and for all purposes.

TECHNICAL FIELD

[0002] The present disclosure relates to drug delivery devices or medicament delivery devices for injecting, delivering, administering, infusing or dispensing substances and/or liquids such as insulin, hormone preparations or vaccines. More particularly, a safety mechanism for a drug delivery device including a needle cover and a lock element that can be in a locking position, includes a lock element that engages the needle cover to prevent the needle cover from being moved out of a needle covering position.

BACKGROUND

[0003] A variety of diseases exist that require regular treatment by subcutaneous administration of a medicament, and a number of drug delivery devices have been developed to support a patient in accurately and controllably delivering an amount of drug in a self-administration process. Delivery devices include injection devices that are removed from the injection site after each medication event or drug delivery process, as well as infusion devices with a cannula or needle that remains in the skin of the patient for a prolonged period of time.

[0004] By way of example, diabetes may be treated by self-administration of insulin or its derivatives with the help of multi-variable-dose insulin injection pens. An injection pen device generally has an elongate device body defining a longitudinal main device axis. An automatic injection device has a motor or a drive spring for biasing a plunger rod and shifting a piston in a container barrel, wherein the drive spring may have to be charged or strained manually prior to injection of a dose. A manually powered delivery drive requires a user to manually provide the energy to move the piston, for instance by applying a distal force component to the injection device.

[0005] The medicament dose to be injected may be manually selected by turning a dosage knob and observing the actual dialed dose from a dose window or display of the insulin pen. A dose is dispensed by inserting the needle into a suited portion of human skin and by moving the piston manually or by pressing a release button of an automatic injection device. Automatic injection devices may include an electronic dose dial mechanism to automatically set a dose.

[0006] Also known is the use of autoinjectors with prefilled syringes. Autoinjectors usually include a body for housing the syringe as well as an automatic drive mechanism in order to move the plunger of the syringe upon actuation of the autoinjector. The drive mechanism typically includes a source of drive, such as a motor or strong spring for moving a transfer element, for example a rod, which acts on the plunger of the syringe.

[0007] For safety and hygiene reasons it is desirable that the needle does not protrude from a housing of the autoinjector with the exception of the time when the needle is used for injection of a medicament. Thus, either the autoinjector moves the needle out of the housing for the injection and back into the housing after injection or the housing provides a needle cover or needle guard which

may be moved to unsheathe the needle for injection and which may be moved back in a needle covering position after the injection.

[0008] The majority of autoinjectors are configured as single use devices which incorporate both the syringe and the drive mechanism in the same housing. Such devices are usually disposable for hygiene reasons.

[0009] Disposable autoinjectors including an electric actuator or an electronic control require a source of energy which is usually in the form of a battery. However, in this case, the autoinjectors should not be disposed in the regular waste, but have to be subjected to special disposal or to recycling, which is an additional burden to the patient. Further, disposing a battery, motor and/or electronics after a single use is a waste of resources and increases the costs of the autoinjector.

[0010] In order to account for a need to handle and dispose the autoinjector parts differently semi-reusable autoinjectors have been developed. Such autoinjectors typically include a reusable drive unit as well as a disposable syringe unit which may be releasably coupled or attached to the drive unit. The drive unit usually includes the drive mechanism and electronics whereas the syringe unit includes the syringe with the injection needle and a needle cover sleeve. The user can thus dispose of the syringe unit when empty or after use and can load the drive unit with a new syringe unit for an upcoming injection.

[0011] Different mechanism for covering and uncovering the injection needle are known in the art. Typically, a cover sleeve or movable sheath can be unlocked and subsequently retracted to expose the needle and prepare the delivery device for an injection.

[0012] WO12038721 A1 discloses a semi-reusable autoinjector with a syringe unit including a syringe and with a drive unit including a torsion spring and a gear to automatically dispense the drug from the syringe. The drive unit includes an opening to accommodate the syringe unit. In a bottom plate in the opening two forwardly projecting lugs are designed to enter into recesses in the syringe unit housing upon insertion of the syringe unit to unlatch a rearward collar of a syringe shield. This allows the syringe shield to move to a retracted position.

[0013] WO 2010/046569 A1 discloses a reusable autoinjector with a syringe unit that can be coupled to a drive unit by a bayonet connection. Once a cam of the drive unit is inserted in a slot in the syringe unit housing a cover sleeve is unlocked and can be moved to unsheathe the needle.

[0014] WO 2021/069106 A1 discloses an autoinjector including a syringe unit and a drive unit. When an axially protruding locking pin of the drive unit is pushed through in a pin opening in the syringe unit, it comes in contact with a syringe holder arm. This results in a deflection of the holder arm, which in turn unlocks a needle cover sleeve and allows it to move in the proximal direction.

SUMMARY

[0015] It is an objective of the disclosure to provide a space-saving design and a safe and reliable release of a needle protection of a drug delivery device.

[0016] According to the disclosure a safety mechanism for a drug delivery device for dispensing a liquid drug from a reservoir through a needle is provided. The safety mechanism includes: [0017] a housing defining a longitudinal axis, the housing may be adapted to accommodate a dispensing mechanism for dispensing the drug and is adapted to accommodate at least partially the reservoir with the drug; [0018] a needle cover movable along the longitudinal axis between a needle covering position, in which the needle is covered by the needle cover and a retracted position, in which the needle cover is moved away from a tip of the needle and the needle is exposed and protrudes from the housing; [0019] a movable lock element that can be in a locking position, in which the lock element may engage the needle cover or a needle cover supporting member or the housing to prevent or block the needle cover from being moved out of the needle covering position towards a retracted position.

[0020] The safety mechanism further includes a trigger member movable along the longitudinal axis relative to the housing. The trigger member may be movable exclusively along the longitudinal axis. The trigger member is operatively and permanently coupled (direct or indirect via

an intermediate member) to the lock element and can actuate the lock element such that a movement of the trigger member along the longitudinal axis causes the lock element to move out of the locking position. Thus, the needle cover is no longer blocked by the lock element and the needle cover can be moved out of the needle covering position towards a retracted position. The delivery device is then ready for an injection as the needle is exposed and accessible. In other words, with an exclusive longitudinal movement of the trigger member (manually or automatically driven) the needle cover can be unlocked and retracted towards an uncovering position.

[0021] In contrast to known approaches that focus on a radial release trigger (in a plane perpendicular to the longitudinal axis), according to the disclosure, the trigger member is movable along the longitudinal axis. As the trigger member does not move in the radial direction the dimensions of the safety mechanism and thus the dimension of the drug delivery device in the radial direction can be reduced compared to known mechanism. Thus, the arrangement according to the disclosure provides for a space-saving design. In view of WO12038721, the disclosure provides at least an alternative design with a trigger member movable relative to the housing.

[0022] The trigger member unlocks the locking element and thus the trigger member may also be referred to as an unlocking member. The trigger member may be, for example, a sleeve or a nut movable relative to the housing. The trigger member may be moved manually by a user force or automatically by an automatic drive such as an electric motor or a preloaded spring member. The trigger member may be arranged inside the housing of the safety mechanism or inside a housing which encloses the safety mechanism and a dose and dispensing mechanism. The trigger member is thus not arranged in an injection device cap or in an external device. The trigger member may be sleeve-shaped and coaxially arranged to a reservoir or syringe inside the housing. This allows for a space-saving design.

[0023] Furthermore, the trigger member can actuate the lock element. That means the lock element can be brought into contact with the lock element either directly or indirectly via an intermediate member. In the latter case the trigger member contacts or engages the intermediate member which in turn contacts or engages the lock element. In this way, a movement of the trigger member is transferred to the lock element to move the lock element out of the locking position towards a release position in which the lock element does not hinder or block the needle cover and the needle cover is movable out of the covering position towards a retracted position.

[0024] Optionally, the longitudinal shifting or movement of the trigger member can be transferred to a radial deflection or movement of the lock element, either directly or indirectly via an intermediate member.

[0025] The lock element is adapted to block or hold the needle cover in the needle covering position. The lock element may be a deflectable, rotatable or shiftable element, for example in form of a ledge, arm, protrusion or knob. Further, the lock element may be a movable element of the needle cover or of a housing part adapted to contact, abut or engage the needle cover.

[0026] The lock element can be in the locking position and may be moved into an unlocking position, intermediate position or end position. If the lock element is in the locking position it may abut a stop surface of the needle cover or of a support or housing element to prevent the needle cover from being moved out of the covering position. If the lock element is not in the locking position it does not prevent the needle cover to be moved out of the covering position. The lock element may be part of the needle cover, of a support element for the needle cover of the housing or any other part of the drug delivery device.

[0027] The needle cover in turn is movable along the longitudinal axis and may be guided within the housing or a member connected to the housing or a member of a reservoir unit attachable to the housing. The movement of the needle cover allows to switch between a safety state in which the needle is covered and the needle does not protrude from the housing and an injection state in which the needle is uncovered and exposes. The needle cover may be implemented as shield or sleeve or sleeve-shaped element adapted to cover or envelop the needle of the drug delivery device the

covering position.

[0028] The drug delivery device may be an injection device, it may be a disposable, reusable or semi-reusable injection pen, an autoinjector or semi-reusable (or semi-disposable) autoinjector or a patch injector applicable onto the skin of the user for the duration of the injection. Reusable injection devices may include at least a part or a unit that can be reused for several injections. Semi-reusable devices include a unit or part that includes a disposable reservoir with the liquid drug which is discarded after an injection or if the reservoir is empty. The semi-reusable devices further include a reusable part typically including a dispensing mechanism.

[0029] Automatic devices or autoinjectors typically include automatic drive means such as an elastic element (for example a preloaded spring) or an electric motor to drive a dispensing member to dispense the liquid drug from the reservoir.

[0030] The safety mechanism may include a counter element and wherein a first guiding surface of the trigger member or the counter element causes the trigger member or the counter element to move the lock element out of the locking position if the trigger member is moved along the longitudinal direction. The counter element may be part of or it may form the above mentioned intermediate member or alternatively the counter element may be a part of the housing of the delivery device.

[0031] If the first guiding surface contacts a contact element as a protrusion, cam or counter surface and if the trigger member is further moved along the longitudinal axis the first guiding surface is pressed onto the contact element the guiding surface slips along the counter element such that the longitudinal movement of the trigger member forces the counter element in a radial direction or in any direction inclined to the longitudinal direction

[0032] The trigger member may include the first guiding surface and the counter element includes a corresponding second guiding surface adapted to contact and engage the first guiding surface. An interaction of the first guiding surface with the second guiding surface causes the trigger member or the counter element to move the lock element out of the locking position.

[0033] The guiding surface may be a first sloped contact surface, it may be arranged on a protrusion, ledge or cam. The second guiding surface may be a second sloped contact surface.

[0034] The first and the second contact surface may be sloped in the same direction or have the same slope angle. The first and second sloped contact surface allow for a reliable actuating of the lock element. Due to the sloped surfaces the longitudinal movement of the trigger member can be reliably transferred to a radial movement or to a movement in an angled direction with respect to the longitudinal axis. In concrete terms, if the first sloped contact surface contacts the second sloped contact surface and if the trigger member is further moved along the longitudinal axis the first sloped surface is pressed onto the second sloped surface thereby causing the two surfaces to slip on each other such that the longitudinal movement of the trigger member forces the counter element in a radial direction or in any direction angled to the longitudinal direction.

[0035] The first and the second sloped contact surfaces may be positioned, for example, relative to the longitudinal axis in an angle of at least 20° or in an angle between 40° and 50° to the longitudinal axis.

[0036] The lock element may be a deflectable arm having an end connected to the needle cover, to the housing or to a holder and the arm has a free end part or end portion adapted to engage a stop surface or counter surface of the needle cover. Thus, if the needle cover is in the covering position the free end part or end portion of the arm engages or abuts the stop surface of the needle cover and thereby prevents a movement of the needle cover out of the covering position towards a retracted position. The lock element in form of a flexible arm allows an easy lock and unlock of the needle cover as the arm can be moved quickly out of the locking position.

[0037] Alternatively, the lock element may be a knob or ledge or the like.

[0038] As mentioned above the trigger member and the lock element are operatively coupled to each other. A movement of the trigger member in a first direction along the longitudinal axis may

cause the lock element to move radially out of the locking position and a movement of the trigger sleeve in a second, counter direction along the longitudinal axis enables the lock element to move radially into the locking position. Thus, the lock element can be moved radially inwards or outwards, respectively depending on the longitudinal movement direction (distally or proximally) of the trigger member.

[0039] The term “move radially” includes movement that have at least a radial movement component. Hence, in case the lock element is a tongue or an arm these elements may swivel or pivot towards and away from the longitudinal axis and such a movement shall be encompassed by the term “move radially”.

[0040] In case the safety mechanism is used in a drug delivery device with an electric motor the motor can be driven forward and backwards to move the trigger member in either direction. Therefore, a simple and reliable locking or unlocking of the needle cover can be provided.

[0041] The drug delivery device may include a reservoir holder for holding the reservoir with the liquid drug. The needle cover may be movable relative to both the lock element and the reservoir holder if the needle cover is moved out of the covering position. In this embodiment the needle cover may be engaged and guided by the reservoir holder.

[0042] The safety mechanism may include a sleeve-shaped support member for holding the reservoir or a reservoir holder encompassing the reservoir. The support member may be arranged proximally or/and outside the trigger member and the support member may have a non-circular cross section. That provides a roll-away protection for the drug delivery device.

[0043] In a first embodiment the safety mechanism includes an intermediate member (or connecting element) which is may be a sleeve and which includes the counter element with a second guiding surface (or a second sloped contact surface). In this embodiment a movement of the trigger member along the longitudinal axis causes the first guiding surface (or first sloped contact surface) of the trigger member to interact with the second guiding surface (second sloped contact surface) of the intermediate member. This moves a portion of the intermediate member in a first radial direction or in an inclined direction thereby causing the lock element to move out of the locking position. The intermediate member may contact or abut the lock element and the lock element is moved by the intermediate member in a radial direction or inclined direction. Thus, the longitudinal movement of the trigger member is redirected to a radial movement of a portion of the intermediate member. This allows for a space-saving design as the trigger member is moved longitudinally and the intermediate member is moved radially. The dimensions of the safety mechanism in a radial direction are thus reduced compared to a mechanism with a radially moving trigger member.

[0044] In this embodiment the trigger member does not directly contact the lock element but the intermediate member is operatively positioned between the trigger member and the lock element. That means the lock element is controlled via intermediate member which allows to transfer a movement of the trigger member by means of a gear ratio or the intermediate member can be used to selectively transfer or not transfer movements of the trigger member to the lock element. Therefore, a sophisticated kinematic control structure can be provided.

[0045] The trigger member with the guiding surface or first sloped contact surface may be exclusively moved along the longitudinal axis where the intermediate member with the second guiding surface or second sloped contact surface is exclusively moved in a radial direction or in a direction angled to the longitudinal axis. The intermediate member may abut, contact or engage the lock element to move it in or out of the locking position.

[0046] The intermediate member may be an arm, cam or protrusion. The intermediate member may be at least partially arranged coaxially inside the trigger member, it may be radially arranged between the lock element and the trigger member. This allows for a space-saving design.

[0047] The arm cam or protrusion may be connected to a sleeve-shaped member forming the intermediate member. In this embodiment the intermediate member has an asymmetric cross

section in a plane perpendicular to the longitudinal axis. That means the intermediate member may have a movable portion or a movable element (in form of the mentioned arm, cam or protrusion) on one side or on a several sides. The movable portion may be adapted to engage the trigger member and/or the lock element. The second guiding surface (second contact surface) may be arranged on or integrally formed in the movable portion of the intermediate member.

[0048] The trigger member may include further a third guiding surface and the intermediate member includes a fourth guiding surface. A movement of the trigger member in a second, counter direction along the longitudinal axis causes the third guiding surface to engage the fourth guiding surface and thereby moving a portion of the intermediate member in a second radial direction allowing the lock element to move from an unlocking or intermediate position into the locking position.

[0049] Therefore, by means of the third and fourth guiding surfaces the lock element can be easily and reliably moved into the locking position (thereby locking the needle cover) and by means of the first and second guiding surface the lock element can be moved out of the locking position (thereby releasing the needle cover).

[0050] As described above with respect to the first and second surface also the third and fourth surface may interact with each other such that the longitudinal movement of the trigger member is transferred to a lock element radial movement or a lock element movement in a direction angled to the longitudinal axis.

[0051] The third guiding surface may be implemented as third sloped contact surface and the fourth guiding surface may be a fourth sloped contact surface.

[0052] Instead of or additionally to the above the intermediate member (or a portion of the intermediate member) may be resilient or elastic such that the intermediate member can be deflected or biased and returns to an initial state due to the elastic properties.

[0053] The intermediate member (or a portion thereof) may be biased by the trigger member movement in the first longitudinal direction when the first guiding surface interacts with the second guiding surface and the intermediate member may return back or deflect back to the initial state by the trigger member movement in the second longitudinal direction when the first and second guiding surfaces disengage from each other.

[0054] The biasing feature enables the intermediate member to switch back or move back to the initial state without additional interaction. This allows to ensure that intermediate member does not unintentionally move the lock element out of the locking position if, for example, the intermediate member is biased in a state in which the lock element is in the locking position.

[0055] The intermediate member may include a radially protruding actuation element in form of a cam, nose, lobe or protrusion. The control element is adapted to interact with and actuate the lock element. Depending on the form and the design of the actuation element a specific movement of the trigger member can be transferred to the lock element. Hence, the design of the actuation element defines how the lock element is actuated and how the lock element is moved by the intermediate member. The actuation element includes a contact portion which is adapted to contact the lock element to move the lock element into or out of the locking position.

[0056] Furthermore, the trigger member includes a control element wherein the first and third guiding surface (third contact surface) are arranged each on an opposite side of the control element in a longitudinal direction. The control element may be implemented as cam or protrusion.

[0057] The control element may protrude, for example, in a radial direction, e.g., in a plane perpendicular to the longitudinal axis. The control element may contact and engage the second and/or fourth guiding surface of the intermediate member if the trigger member is moved along the longitudinal direction. The control element allows a simple controlling and transfer of movement from trigger member to the intermediate member.

[0058] The second and fourth guiding surface may be arranged on an intermediate control element (cam or protrusion), which may be directed in a radial direction too. Therefore, the interaction

between the trigger member and the intermediate member is implemented via control element and intermediate control element again allowing for a space-saving design.

[0059] The safety mechanism according to the disclosure includes in an embodiment a cap releasably attached at a distal end of the needle cover in an initial state or unused state of the drug delivery device. In this embodiment the needle cover or the cap includes a resilient and deflectable element (tongue, arm, protrusion or the like) holding the cap on the needle cover in a non-deflected state. A cap actuation element is operatively (direct or indirect) coupled to both the trigger member and the deflectable element. A movement of the trigger member in the longitudinal direction causes a movement of the cap actuation element which in turn causes a change of the deflectable element from the non-deflected state to a deflected state thereby releasing the cap from the needle cover. That means after the deflectable element is deflected the user can remove the cap from the needle cover to prepare the delivery device for an injection.

[0060] The deflectable element may be separate from the needle cover or from the cap or it may be integrally formed (monolithic) in the needle cover or in the cap. The deflectable feature allows to connect the cap easily by a releasable snap-fit connection.

[0061] In an embodiment the intermediate member includes the cap actuation element (e.g., the cap actuation element may be integrally formed in the intermediate member) and the trigger member includes further a first cap surface and the intermediate member includes a second cap surface which are both sloped surfaces, wherein the movement of the trigger member in the first direction along the longitudinal axis causes the first cap surface to engage the second cap surface thereby moving the intermediate member (or a portion thereof) in a first radial direction which causes the deflectable element to deflect.

[0062] The trigger member may include further a third cap surface and the intermediate member includes a fourth cap surface which are both sloped surfaces. A movement of the trigger member in a counter second direction along the longitudinal axis may cause the third cap surface to interact with the fourth cap surface thereby moving the intermediate member (or a portion thereof) in a second radial direction allowing the deflectable element to deflect back in its initial position.

[0063] The intermediate member includes in this embodiment not only an actuation element for moving the lock element but also a cap actuation element, for example in form of a protrusion, nose, lobe or cam for moving the deflectable element. That means the intermediate member provides two functions: actuating the lock element (for holding or releasing the needle cover) and actuating the deflectable element (for holding or releasing the cap). Hence, two function can be provided with one single member. For that purpose, the intermediate member may include two independently deformable portions, a first deformable portion for actuating the lock element and a second deformable portion for actuating the deformable element for the cap.

[0064] In a second embodiment of the safety mechanism according to the disclosure the housing includes the counter element. Thus, in contrast to the above mentioned first embodiment no intermediate member is present and the counter element is integrally formed in the housing or in a member fixedly connected to the housing.

[0065] The trigger member may include a body and a deflectable portion including the first guiding surface (sloped contact surface) and movement of the trigger member body in the first direction along the longitudinal axis causes the first guiding surface (sloped contact surface) to interact with the housing thereby moving the deflectable portion of the trigger member in a first radial direction or in a direction inclined with respect to the longitudinal axis thereby engaging and moving the lock element out of the locking position towards a non-locking position. Hence, the trigger member directly contacts and engages the counter element without any intermediate member.

[0066] Optionally, the housing includes a second guiding surface adapted to interact with the first guiding surface.

[0067] As described above with respect to the first embodiment when the lock element is moved out of the locking position the needle cover can be moved out of the covering position and thus

exposing the needle. If the housing features the second guiding surface a simple design is provided as no further member or part is required. The first and second guiding surfaces may be a first and second sloped contact surface.

[0068] The deflectable portion of the trigger member may be, for example, implemented as arm, tongue or deflectable protrusion. The deflectable portion may be essentially moved in a first radial direction or in a direction angled to the longitudinal axis by the longitudinal movement of the body of the trigger member.

[0069] In second embodiment the safety mechanism may include a cap releasably attached to a distal end of the needle cover in an initial state, shipping state or unused state of the delivery device. The cap or the needle cover includes a deflectable element holding the cap on the needle cover, it may be in a non-deflected state. A movement of the needle cover in the longitudinal direction out of the covering position towards a retracted position causes a change of the deflectable element from the non-deflected state to a deflected state thereby releasing the cap from the needle cover.

[0070] In an embodiment an automatic movement of the needle cover and cap relative to the housing releases the cap from the needle cover.

[0071] The cap can thus be hold on the needle cover by a releasable snap-fit connection.

Furthermore, a simple longitudinal movement of the needle cover may release the connection and the cap can be removed and the delivery device is ready for an injection.

[0072] The disclosure relates further to an injection device for dispensing the liquid drug from the reservoir. The injection device includes a plunger rod and an automatic drive for driving the plunger rod to move a piston inside the reservoir to dispense the liquid drug form the reservoir through the injection needle or cannula. The injection device further includes the safety mechanism as described above.

[0073] The injection device may be, for example, a manual user-driven or automatic injection pen, an autoinjector including a spring element or motor for automatically driving the plunger rod or a semi-reusable injection device or semi-reusable autoinjector. Such reusable devices include a reusable drive unit and a disposable reservoir unit or syringe unit which is releasably attachable (for example by a thread or bayonet connection) to the drive unit.

[0074] The automatic drive may be adapted to rotate a drive member of the drive unit, where the drive member is operatively coupled (direct or indirect) to both the plunger rod and the trigger member, wherein rotation of the drive member by the automatic drive in a first rotational direction causes the plunger rod and the trigger member to be moved in the first direction along the longitudinal axis and rotation of the drive member in a second, counter rotational direction causes the plunger rod and the trigger member to be moved in the second direction along the longitudinal axis.

[0075] Hence, with one single automatic drive the plunger rod can be moved back and forth and the trigger member can be moved back and forth at the same time. That allows to lock and unlock the needle cover (via lock element) and, if any, lock and release a cap on the cover sleeve (via intermediate member, if any) and with the same automatic drive the plunger rod can be driven to dispense the drug form the reservoir. That means the automatic drive can be used to rotate the drive member to move the trigger member before an injection to lock or unlock the cover member and potentially the cap and subsequently the automatic drive can be used to dispense the drug during an injection or for preparation steps.

[0076] The drive member may be in treaded engagement with the plunger rod, such that the plunger rod is exclusively shifted along the longitudinal direction when the drive member is rotated. The trigger member may be directly or indirectly coupled to the plunger rod such that a longitudinal movement of the plunger rod in the first direction causes to move the trigger member in the longitudinal direction in the first direction and movement of the plunger rod in the second longitudinal direction causes to move the trigger member in the second longitudinal direction.

[0077] The injection device further may include a reservoir unit (or syringe unit) and a drive unit. The reservoir unit may be releasably attachable to the drive unit. The reservoir unit includes a reservoir for the liquid drug (e.g., a cartridge, a syringe or a container), the needle cover and the lock element and the drive unit includes the housing, the automatic drive, the plunger rod, the trigger member and, if any, the intermediate member.

[0078] The reservoir unit may be disposable and the drive unit may be reusable. That means the injection device includes two units allowing to establish a semi-reusable concept which reduces the waste. In contrast to conventional injection devices the semi-reusable device does not have to be discarded completely but only the reservoir unit can be disposed after use or if the drug is fully dispensed out of the reservoir. The drive unit can be reused for several injections.

[0079] The reservoir unit is releasably attachable to the drive unit. That means the two units can be detached and reattached without destroying elements. Non-limiting examples of releasable connections are a snap-fit, a bayonet connection or a thread.

[0080] In an embodiment the reservoir unit includes a radially extending protrusion or radially extending rim portion and the drive unit includes a radially deflectable and flexible holding arm or clamping arm adapted to engage the protrusion or rim portion such that the syringe unit is held by the drive unit via snap-fit connection.

[0081] The drive unit may have additionally a lock to lock the snap-fit connection such that the syringe unit cannot be removed from the drive unit if the lock is activated. The lock may be implemented by the above described lock element or/and by a separate lock element preventing, for example, a radial movement of the holding arms and thereby locking the arms relative to the protrusion or rim portion.

[0082] The reservoir inside the reservoir unit may be a prefilled syringe with a non-releasable connected injection needle at the distal end thereof.

[0083] Upon attachment the reservoir unit may be held by the drive unit relative to the housing such that a movement of the reservoir unit in a dispensing direction is prevented.

[0084] The disclosure further refers to a method for a safety mechanism for an injection device for dispensing a liquid drug from a reservoir through a needle. The method includes the steps of [0085]

a. Providing an injection device with a housing defining a longitudinal axis; [0086] b. Moving a trigger member inside the housing along the longitudinal direction [0087] c. Moving a lock element, by the trigger member as the trigger member is operatively coupled to the lock element, out of a locking position in which the lock element engages a needle cover to prevent the needle cover from being moved out of a needle covering position, [0088] d. Releasing, by the movement of the lock element, the needle cover from the covering position and moving the needle cover out of the needle cover position towards a retracted position in which the needle is exposed.

[0089] A further aspect of the disclosure relates to an injection device for injection of fluid medicaments into a patient's tissue. Examples of possible injection devices are injection pens, autoinjectors, patch injection devices or the like. The injection device according to the disclosure includes electronics, with at least an electronic controller and a near field communication unit in functional connection to the controller. The injection device according to the disclosure further includes a cartridge assembly. The cartridge assembly has to be construed broadly. In one possible arrangement the cartridge assembly is mainly constituted by the cartridge as such, where the cartridge may be formed by a prefilled syringe, ampule or the commonly used carpule. In another arrangement the cartridge assembly may be constituted by a cartridge unit, e.g., including a housing, the cartridge as such, a needle protection sleeve, a needle assembly, and/or further constituents. It has to be understood that the cartridge assembly shall not be construed as a fully functional injection device of its own. So consequently, a driving arrangement shall not be part of the cartridge assembly even though particular elements of a driving arrangement may be located in the drive arrangement. The cartridge assembly includes in any case a near field communication readable item fixedly attached to the cartridge assembly. The near field communication readable

item, in a simple design according to the disclosure, is a RFID or NFC (smart) tag (according to ISO/IEC 14443A/B, ISO/IEC15693, ISO/IEC18000-3, and/or according to any other suitable standard). Such tags may store information which may be read by a reading device. For example a tag according to the disclosure may store information regarding the medicament contained in the cartridge, the manufacturing date of the cartridge assembly and/or the medicament as well as any other information considered useful in the context of a therapy or in relation to the therapy. However, such tags may store a limited amount of data only, typically ranging from 48 Bytes to 1 Mbyte. For example, the NTAG213 by NXP (www.nxp.com) includes a user storage of 144 Bytes. According to the disclosure the near field communication unit of the injection device is capable of acquiring the data stored on the near field communication readable item. The cartridge assembly may be fixedly attached to the injection device. For example, the actual injection device may be manufactured at a first site, a prefilled syringe including the near field communication readable item (tag) as the cartridge assembly may be manufactured at a second site, and the end assembly, i.e. the mounting of the prefilled syringe, may take place at a third site. With the tag associated with the prefilled syringe it is not necessary to activate the injection device electronics during assembly, as all the necessary information may be obtained by the injection device at any point in time particularly also after assembly which simplifies the end assembly process significantly.

Alternatively, the cartridge assembly may be releasably attachable to the injection device. In any case, if the cartridge assembly is attached to the injection device, the near field communication readable item and the near field communication unit are geometrically arranged at a distance to each other which allows for the stored information to be transferred from the item to the unit. The person skilled in the art is aware of appropriate geometrical arrangements for near field communication arrangements. The near field communication unit according to the disclosure can be used in two modes. In a first mode, the near field communication unit is used to read information from the near field communication readable item. In the second mode, the near field communication unit is used to provide information to another device. The other device is a separate device, which includes a near field communication reading device. For example, the near field communication unit may read information from the near field communication readable item, provide the read information to the controller and subsequently change mode to provide at least a fraction of the read information and/or further data to the separate device. The information provided may be based on the read information or on information created by the controller.

[0090] In advantageous embodiments the near field communication device may use one single antenna for sending and receiving data.

[0091] In advantageous embodiments the injection unit includes only one wireless communication means, namely the near field communication unit, as described above.

[0092] In an embodiment the other device as described above may act as a display device for the injection device, rendering moot the need for an alphanumeric display, like an LCD or an OLED display, arranged on the injection device.

[0093] In a further developed example, the near field communication device of the injection device may be used not only to read information from the cartridge but also from the other device, which is e.g., a smart phone. For example the smart phone may receive recall information regarding the medicament used in the injection device and transfer information via near field communication containing e.g., a list of recalled lot number to the injection device. In case the lot number of medicament currently used in the injection device matches an entry of the list, the injection device may alert the user and/or block the injection device. The alert may be first transferred to the smart phone and be displayed on the display of the smart phone. Another possibility of a data transfer from the other device (smart phone) to the injection device might relate to the transfer of a kind of therapy plan from smart phone to injection device.

[0094] The controller may monitor operation of the injection device according to the disclosure or it may control at least part of the operation of the injection device according to the disclosure.

[0095] One aspect of the disclosure relates to an injection device as described where near field communication means near field communication (NFC) according to the standard ISO/IEC 18092 and its normative references or any of related standards by the IEEE organization or related.

[0096] One aspect of the disclosure relates to a system combining the injection device according to the disclosure and the separate device. The separate device may be a smart phone, a cellular phone, a personal digital assistant (PDA) like a palm pilot, a tablet computing device or any computing device with a near field communication reading device, like an NFC or a RFID reading device. The separate device runs AndroidOS, iOS by Apple, Windows operating system, Linux operating system, MacOS operating system, or any related operating system capable of operating the near field communication reading device.

[0097] In one aspect of the disclosure the information provided by the near field communication unit of the injection device to the separate device is data relating to state information of the injection device or error information, both gathered by the controller. The error information may include error messages from the injection device as such, like operation failure, battery low state or the like. The error information may further or alternatively contain error information related to information acquired from the near field communication readable item, e.g., reached expiration date of the medicament, wrong medicament identification, or the like.

[0098] In one aspect of the disclosure the injection device according to the disclosure is a modular injection device. The modular injection device includes at least a reusable module and a disposable module which may be releasably attached to the reusable module. The reusable module contains at least the controller and the near field communication unit. Disposable and reusable modules are attached to each other by snap, bayonet, threaded connections or the like. The disposable module includes at least the cartridge assembly. After use or depletion of the cartridge, the disposable module is removed from the reusable module and disposed. A new disposable module may then be attached to the reusable module for further use.

[0099] In a more detailed example of the modular injection device according to the disclosure the injection device may be an autoinjector. The disposable unit is a cartridge unit. The reusable unit has a housing which houses at least a driving arrangement configured to force medicament out of a cartridge, electronics with a controller, and a near field communication unit. In, at, or on the housing a user interface is arranged which may be in functional communication with the controller. The cartridge unit includes a housing in which the cartridge with the fluid medicament is arranged. At a surface of the housing of the cartridge unit a near field communication readable item, for example a NFC tag, is arranged. The near field communication readable item stores information regarding the medicament, such as the type of medicament, the manufacturing date, information regarding shelf life and the like. Optionally, the tag may further include a temperature sensing element for logging of at least a maximum storage temperature or a humidity sensor.

[0100] The reusable unit's housing is configured to accommodate the cartridge unit at least partly within the housing with typically having the delivery end of the cartridge reaching out of a distal end of the reusable unit. The cartridge unit may releasably attached within the housing of the reusable unit for use. As the cartridge unit is accommodated in the reusable unit, the near field communication readable item of the cartridge unit gets located in the very proximity of the near field communication unit of the reusable unit. Driven by the controller, the near field communication may be used to read the information stored in the near field communication readable item. The read information may be used by the controller to influence operation of the injection device. Moreover, the information may be stored on the controller or a storage unit attached to the controller.

[0101] According to the disclosure the near field communication unit of the reusable unit of the injection device is not only configured to receive information but also to transmit information to other devices. In order to enable data transmission the controller is configured to set the near field communication unit into the so-called card emulation mode. Once in this mode the near field

communication unit simulates a nearfield communication device from which information may be read, similar to the near field communication readable item as described above. One possibility for another device which reads or receives information from the injection device may be a smart phone. The smart phone may be used to monitor a therapy and/or to give guidance to a person using the injection device.

[0102] In a further aspect the driving arrangement of the injection device includes an electric or electronic drive train including an element movable along an axis of the injection device which may be an autoinjector to force medicament out of the cartridge. For example, the movable element may be a piston rod applying force on a moveable piston in the cartridge, which in an example may be a prefilled syringe. The movable element may be driven by an electric motor, like a stepper motor, a DC motor, or piezoelectric motor. A gearing setup may interconnect the motor and the movable element.

[0103] In a further aspect, electrical and/or electronic components of an injection device according to the disclosure are energized through an energizing element like a battery, an accumulator or a super cap, wherein the energizing element of the injection device is disposed within a housing or a housing part of the injection device. The energizing element may be fixedly and non-removably disposed in the injection device. In an alternative the energizing element may be removably disposed in the injection device, either for recycling purposes or for exchange during use of the injection device.

[0104] In a further aspect, an injection device according to the disclosure includes a user interface. The user interface is operationally connected to the controller of the injection device. In a simple example the user interface consists of one or more LEDs (light emitting diodes), which may emit signals provided by the controller and meaningful to a person using the injection device. The signals may relate to a state of the injection device and/or to information received through near field communication from the cartridge or the cartridge unit.

[0105] In a more sophisticated example the user interface includes an LCD or OLED Display capable of displaying more comprehensive information regarding a device state and/or information regarding the cartridge.

[0106] Various described aspects in this disclosure relate to near field communication. Near field communication with regards to the present disclosure shall be construed in its broadest possible sense as a wireless communication method using radio frequency electro-magnetic radiation to transmit information over a distance of less than one centimeter to several centimeters.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0107] The subject matter of the disclosure is explained in more detail in the following text with reference to exemplary embodiments which are illustrated in the attached drawings, in which:

[0108] FIG. 1 depicts an isometric view of a semi-reusable autoinjector, according to the disclosure;

[0109] FIG. 2 depicts an exploded view of the syringe unit of the autoinjector from FIG. 1;

[0110] FIG. 3 depicts a sectional view of the drive unit of the autoinjector;

[0111] FIG. 4 depicts the drive unit with an inserted syringe unit;

[0112] FIG. 5 depicts a sectional view of the autoinjector where the syringe unit is locked to the drive unit;

[0113] FIG. 6a depicts a syringe unit according to a second embodiment;

[0114] FIG. 6b depicts the syringe unit of FIG. 6a but without cap;

[0115] FIG. 7 depicts a sectional view of a drive unit of the second embodiment;

[0116] FIG. 8 depicts a perspective view of a gripper sleeve of the drive unit of the second

embodiment;

[0117] FIG. **9a** depicts the drive unit with an inserted syringe unit according to the second embodiment;

[0118] FIG. **9b** depicts a sectional view of control cams for unlocking a cover sleeve and cap cams for unlocking a cap;

[0119] FIG. **10** depicts the cap cams in a locking state;

[0120] FIG. **11** depicts the control cams as well as the cap cams in a locking state;

[0121] FIG. **12** depicts the control cams and the cap cams during an injection;

[0122] FIG. **13** depicts the autoinjector after release of the syringe unit and

[0123] FIG. **14** depicts a third embodiment of the autoinjector;

[0124] FIG. **15** depicts a syringe unit configured to be inserted into a reusable drive unit of the injection device which may communicate with a smart phone;

[0125] FIG. **16** depicts the injection device with its outer distal housing removed revealing the near field communication unit; and

[0126] FIG. **17** depicts a longitudinal cross section of the injection device showing the syringe unit with the prefilled syringe. The further elements present in the injection device except the housing element are omitted.

[0127] The reference symbols used in the drawings, and their primary meanings, are listed in summary form in the list of designations. In principle, identical parts are provided with the same reference symbols in the figures.

DETAILED DESCRIPTION

[0128] The term “injection device” or “injector” refers to a device that is removed from the injection site after each medication event or drug delivery process. This is in contrast to the term “infusion device” which refers to a device with a cannula or needle that remains in the skin of the patient for a prolonged period of time, for example, several hours.

[0129] In the present context, the terms “substance”, “drug”, “medicament” and “medication” are to be understood to include any flowable medical formulation suitable for controlled administration through a means such as, for example, a cannula or a hollow needle, and includes a liquid, a solution, a gel or a fine suspension containing one or more medical active ingredients. A medicament can be a composition including a single active ingredient or a pre-mixed or co-formulated composition with more than one active ingredient present in a single container. Medication includes drugs such as peptides (e.g., insulin, insulin-containing drugs, GLP-1 containing drugs or derived or analogous preparations), proteins and hormones, active ingredients derived from, or harvested by, biological sources, active ingredients based on hormones or genes, nutritional formulations, enzymes and other substances in both solid (suspended) or liquid form but also polysaccharides, vaccines, DNA, RNA, oligonucleotides, antibodies or parts of antibodies but also appropriate basic, auxiliary and carrier substances.

[0130] The term “distal” refers to the direction or the end of the drug delivery device carrying an injection needle or an injection cannula, whereas the term “proximal” is meant to refer to the opposite direction or end pointing away from the needle or cannula. In the following description the radial direction is meant to refer to a direction perpendicular to the longitudinal direction or longitudinal axis. The latter is defined by the housing and refers to the direction in which the housing (and the delivery device) has the longest dimension. In the present description the term “distal” refers to the side where the injection needle is located. This is on the left hand side in the FIGS. **1** to **13**. The term “proximal” refers to the opposite side or rear end of the device and is on the right hand side in FIGS. **1** to **13**.

[0131] First embodiment: FIG. **1** shows a perspective view of a drug delivery device of a first embodiment of the disclosure in form of a semi-reusable autoinjector **1**. The autoinjector **1** includes a disposable syringe unit **2** and a reusable drive unit **3**. As shown in FIG. **1** the drive unit **3** has an elongated housing defining a longitudinal axis and includes a button **4** at its proximal end. In the

following, the structural features of the syringe unit and the drive unit will be described in detail. Subsequently, the function of the semi-reusable autoinjector **1** will be explained.

[0132] FIG. **2** depicts an exploded view of the syringe unit **2**. As shown in FIG. **2** the syringe unit **2** includes a syringe holder **30**, a cover sleeve **20** coaxially arranged around the syringe holder **30** and movable relative to the syringe holder along the longitudinal axis and a prefilled syringe **15** including a liquid drug and a rigid needle shield **17** on the injection needle. The cover sleeve includes a label **25** wrapped around the cover sleeve **20**. Further, a cap **10** is arranged on a distal end to cover the injection needle. The syringe holder **30**, the cover sleeve **20** and the cap **10** are individually injection-molded plastic parts. The cap **10** may include additional metallic parts.

[0133] The prefilled syringe **15** with a barrel made of glass is rotationally and axially fixed inside the sleeve-shaped syringe holder **30**. For that purpose, the syringe holder **30** includes clamping elements pressing on an outer surface of the syringe barrel and a distal shoulder of the syringe abuts a bearing surface **34** (see FIG. **4**) inside the syringe holder **30**. As shown in FIG. **2** the syringe holder **30** further features longitudinal guiding rails **31** on its outside for the cover sleeve **20** such that the cover sleeve **20** can be shifted exclusively in the longitudinal direction relative to the syringe holder **30**. Furthermore, the syringe holder **30** includes two radially deflectable lock elements in form of locking arms **32** arranged on opposite sides of the syringe holder **30**. In an initial state the locking arms **32** are in a non-deflected state and in a locking position and prevent a movement of the cover sleeve **20** relative to the syringe holder **30** in a proximal direction. Hence, the cover sleeve **20** shields or covers the injection needle of the syringe **15** in a covering position. For that purpose, a free end of the locking arms **32** abut stop surfaces **24** inside the cover sleeve **20**. As will be described below once the locking arms **32** are deflected radially inwards the cover sleeve **20** can be pushed proximally. The syringe holder **30** further includes two oppositely arranged and radially protruding ledges **33** at a proximal end of the syringe holder **30** for engagement with clamping arms **61** (FIG. **4**) inside the drive unit **3**.

[0134] The cover sleeve **20** has a non-circular cross section in a plane perpendicular to the longitudinal axis. That means two sides of the cover sleeve are flattened. The cover sleeve includes a first lateral opening **21** for access to the locking arms **32** and a proximal second opening **22** for access to the ledges **33** from outside. The label **25** wrapped around the cover sleeve **20** includes a machine-readable tag in form of a RFID or QR code **26** identifying the syringe unit **2**.

[0135] At a distal end of the cover sleeve **20** the cap **10** is releasably attached by a snap-fit connection. The cap **10** includes on two sides two deflectable elements or tongues **11** and on each side a protrusion engaging in a corresponding indentation **23** in the cover sleeve **20** for the snap-fit connection. Alternatively, the cap **10** may have an ellipse shape in a plane perpendicular to the longitudinal axis wherein inside the ellipse form the protrusions engage the indentation in the cover sleeve. The ellipse can be deformed to a circular shape to release the protrusions from the indentation **23**. The release of the snap-fit will be described below with respect to the de-capping of the syringe unit **2**. The cap **10** includes in its inside a sleeve-shaped holder **12** (FIG. **2**) connectable to the rigid needle shield (RNS) **17** of the syringe **15** such that the RNS **17** is removed with the cap **10** if the cap is released from the needle cover.

[0136] The syringe **15** inside the syringe holder **30** has a barrel made of glass and inside the barrel a movable piston **16** (FIG. **4**). An injection needle **18** (FIG. **4**) is non-releasably connected to a distal end of the syringe barrel. In an unused state the injection needle **18** is covered by the RNS **17**. The latter in turn is axially and radially engaged or engageable by the holder **12** in an assembled or ready-to-use state of the syringe unit.

[0137] FIG. **3** depicts a sectional view of a first embodiment of the drive unit **3**. The cut of the sectional view runs along the longitudinal axis of the device. The drive unit **3** includes a housing **80** including an outer housing cover and inside the cover a sleeve mechanism **83** including a radial wall **84** that separates the drive **3** unit in a distal portion and a proximal portion. The mechanics sleeve **83** is immovably connected to the housing **80**. Inside the housing **80** and distally the

separating wall **84** is a distal housing insert **81** whereas proximal the wall **84** a proximal housing insert **85** is located. The distal and proximal housing inserts **81**, **85** are immovably connected to the housing **80**. Furthermore, the drive unit **3** includes an opening on its distal side adapted to accommodate a proximal portion of the syringe unit **2**.

[0138] Inside the distal housing portion and guided by the distal housing insert **81** a gripper sleeve **60** is arranged and movable relative to the housing **80** along the longitudinal axis. Coaxially to the gripper sleeve **60** a trigger sleeve **50** is arranged. Again inside the trigger sleeve **50** a cover sleeve connector **45** is located and biased in distal direction by a cover sleeve spring **46**. The spring abuts on its distal end the cover sleeve connector **45** and on its proximal end the radial wall **84** of the sleeve mechanism **83**. A sleeve-shaped syringe connector **47** is coaxially arranged inside the cover sleeve connector **45** and biased distally by a syringe connector spring **48** coaxially inside the cover sleeve spring **46** and proximally supported by the wall **84**. The spring **46** ensures that the syringe connector **45** pushes the syringe **15** of an inserted syringe unit **2** in distal direction.

[0139] The proximal housing portion accommodates the proximal housing insert **85** which supports and guides a sleeve-shaped trigger sleeve connector **51** movable along the longitudinal axis, a drive assembly with an electric motor **91**, a pinion **92** and a gear **93** connecting the motor **91** with a threaded rod **41**, a battery **94**, the treaded rod **41** and a plunger rod **40** threadedly connected to the threaded rod **41**. The proximal housing portion further accommodates an electronic module with a controller configured to control the electronic motor **91** and providing information to the user via a display or LEDs or a communication module. The trigger sleeve connector **51** is fixedly and immovably connected to the trigger sleeve **50** and distally pressed onto a proximal end of the plunger rod **40** by a trigger sleeve spring **53**. The plunger rod **40** non-rotatably guided by the sleeve mechanism **83** and includes on its distal end a flange **49** adapted to engage the piston **16** (see FIG. **4**) during dispensing.

[0140] Inside the proximal housing portion is further an axially arranged detection pin axially movable by the cover sleeve connector **45**. The detection pin is connected to and interacts with a TMR position sensor. The presence of a syringe unit **2** inside the drive unit housing **80** is thus detectable via detection pin and the position sensor which provides a corresponding signal to the controller of the electronic module.

[0141] The gripper sleeve **60** includes two axially extending and deflectable clamping arms **61** having a clamp **62** at their free distal end adapted to accommodate the syringe holder ledges **33**. The clamping arms **61** can be deflected radially outwards allowing the clamps **62** to snap onto the ledges **33** of the syringe holder **30** when the syringe unit is inserted into the opening in the drive unit **3**. On its outside the gripper sleeve **60** includes splines running in corresponding longitudinal grooves in the distal housing insert **81** guiding the gripper sleeve **60** during an axial movement relative to the housing **80**. Furthermore, each clamping arm **61** provides a support for an axially aligned gripper sleeve spring **64** which biases the clamping arms **61** in a holding position.

[0142] The trigger sleeve **50** is axially guided by the gripper sleeve **60** as well as by the distal housing insert **81** and shiftable relative thereto. For this purpose, the trigger sleeve **50** includes longitudinal guiding rails engaging longitudinal grooves in both the distal housing insert **81** and the gripper sleeve **60**. As it can be seen in FIG. **3** the trigger sleeve **50** includes oppositely arranged trigger arms **54** which include on their proximal free end **56** a guiding surface or sloped contact surface **55**. The trigger arms **54** are deflectable radially inwards upon contact of the contact surface **55** with a counter guiding surface or counter sloped contact surfaces **82** of the distal housing insert **81**. As mentioned above the trigger sleeve **50** is further immovably connected to the trigger sleeve connector **51** in the proximal housing portion. As the trigger sleeve connector **51** is biased by the trigger sleeve spring **53** the trigger sleeve **50** is biased too in distal direction. The trigger sleeve connector **51** in turn is connected to the plunger rod via ledges **52** which engage a proximal shoulder of the plunger rod **40** such that an axial movement of the plunger rod **40** in proximal direction moves also the trigger sleeve **50** proximally.

[0143] In the state shown in FIG. 3 the drive unit 3 is ready to be loaded with a syringe unit 2. The controller displays the current state to the user via display by showing a message like “Syringe Unit Insertion” or via green or red LEDs.

[0144] In order to prepare the autoinjector 1 for an injection the user attaches a syringe unit 2 to the drive unit 3. The electronics in the drive unit 3 are switched from an inactive state or sleep mode to an active mode. Alternatively, the user can press the button 4 to activate the electronics.

[0145] FIG. 4 depicts a sectional view of the drive unit 3 with an inserted syringe unit 2. The syringe unit 2 has been inserted with its proximal end portion into the distal opening of the drive unit housing 80. In the state shown in FIG. 4 the proximal ledges 33 of the syringe holder 30 are snapped into recesses of the clamps 62 of the gripper sleeve clamping arms 61. The syringe unit 2 is thus held in the drive unit 3 by the clamping arms 61. However, in the shown state in FIG. 4 the syringe unit 2 can still be pulled out of the drive unit housing by overcoming a holding force. Hence, if the user pulls the syringe unit 2 in a distal direction the syringe holder 30 with the syringe 15 are firstly moved in distal direction relative to the cover sleeve 20 until stop and subsequently the cover sleeve 20 is moved distally too. This brings inclined release surfaces 27 of the cover sleeve 20 into contact with inclined counter surfaces 65 of the clamping arms 61. If the cover sleeve 20 is further pulled in distal direction the clamping arms 61 are deflected radially outwards via release surfaces 27 and counter surfaces 65. The ledges 33 of the syringe holder 30 are therefore released from the clamping arms 61 and the syringe unit 2 can be removed from the drive unit 3. out of the drive unit 3 the clamping arms 61 may deflect radially outwards and release the ledges 33.

[0146] That means in the state shown in FIG. 4 the syringe unit 2 is held inside the drive unit 3 but not yet locked inside the drive unit 3.

[0147] FIG. 5 depicts the autoinjector 1 in a locked state without cap. When the syringe unit 2 is inserted as described above the syringe unit 2 moves the detection pin and thus the position sensor detects the movement and automatically sends a signal to the controller in the electronic module which detects the presence of the syringe unit 2 inside the drive unit housing. A code reader in form of an NFC reader or QR code reader inside the drive unit 3 reads the RFID or QR code 26 of the syringe unit 2. Subsequently, the controller of the drive unit sends the read medication information from the read code 26 to an external device to verify the drug or medication with predefined information. Alternatively, the controller compares the read medication information from the code 26 with predefined information stored in a memory inside the drive unit 2.

[0148] In case of successful medication verification the controller receives an enable signal (either from the external device or from the internal comparison). The controller then controls the electric motor 91 to rotate the threaded rod 41 to move the trigger sleeve 50 a short distance of several millimeters in proximal direction. Such a trigger sleeve 50 movement is achieved by a kinematic chain as described as follows.

[0149] The motor 91 rotates the pinion 92 and the gear 93 transfers the rotation to the threaded rod 41 which in turn moves the non-rotating plunger rod 40 in proximal direction. As the trigger sleeve connector 51 is connected via its ledges 52 to a proximal shoulder of the plunger rod 40 the latter causes the trigger sleeve connector 51 to move and thus the trigger sleeve 50 is shifted in proximal direction too.

[0150] The proximal movement of the trigger sleeve 50 brings the sloped contact surfaces 55 of the trigger arms 54 into contact with the counter sloped contact surfaces 82 of the distal housing insert 81. As the two sloped surfaces 55, 82 slide along each other the trigger arms 54 are forced to deflect radially inwards when the trigger sleeve 50 is further moved in proximal direction. The free ends 56 of the deflected trigger arms 54 in turn come into contact with the locking arms 32 of the syringe holder 30. Subsequently, the trigger arms 54 deflect the locking arms 32 radially inwards and thus move the locking arms 32 out of a locking position. As a consequence, the locking arms 32 no longer engage the stop surface of the cover sleeve 20 and thus allow for the cover sleeve 20

to move out of a covering position and move inside the housing **80** in proximal direction relative to the syringe holder **30** and relative to the drive unit housing **80**.

[0151] At the same time the radially inwards deflected trigger sleeve arms **54** abut a distal stop surface of the syringe holder **30** and thereby lock the syringe holder **30** relative to the trigger sleeve **50** such that a movement of the trigger sleeve **50** is transferred to the syringe unit **2**. Thus, in this state (FIG. 5) the syringe unit **2** can no longer be pulled out of the drive unit **3**.

[0152] The controller displays the locked state to the user via display. As in this state the cap is still mounted onto the distal end of the cover sleeve **20** the next step is the de-capping which may be started by pressing the button **4**. The controller then drives the motor **91** to move additionally the trigger sleeve **50** a short distance of 1 to 5 mm in proximal direction and immediately the same distance back in distal direction. As the trigger arms **54** fix the syringe unit **2** relative to the trigger sleeve **50** the syringe unit **2** (and the gripper sleeve **60**) are moved back and forth too. The proximal movement of the syringe unit **2** with the cap **10** relative to the drive unit housing **80** compresses the trigger sleeve spring **53**. In FIG. 5 it can be seen the trigger sleeve connector **51** was moved a distance in proximal direction (see distance between ledges **52** and proximal end of sleeve mechanism **83**). The proximal movement of the syringe unit **2** relative to the housing **80** further causes the deflectable cap tongues **11** (FIG. 2) of the cap **10** to abut against a triangle-shaped protrusion **86** (FIG. 3) on a distal end of the drive unit housing **80**. As a consequence, if the syringe unit **2** is further moved into the drive unit housing **80** the protrusion **86** deflects the cap tongues **11** and thereby releases the snap-fit connection of the cap **10** with the cover sleeve **20**. Consequently, the cap **10** is wiped off. This leads to the state shown in FIG. 5. The controller moves then motor **91** in counter direction to move the plunger rod **40** back the short distance of 1 to 5 mm but without unlocking the syringe unit **2**. The syringe unit **2**, the trigger sleeve **50** and the gripper sleeve **60** are moved back due to the spring force of the trigger sleeve spring **53**.

[0153] The autoinjector **1** is now de-capped and ready for an injection. The controller displays a corresponding notification on the display.

[0154] As a next step the user places the distal end of the cover sleeve **20** onto an injection site and pushes the autoinjector **1** towards the injection site. This causes the cover sleeve **20** to move proximally from the distal covering position into the drive unit housing **80** (push on skin) and compresses the cover sleeve spring **46** which biases the cover sleeve **20** in distal direction via cover sleeve connector **45**. Upon push on skin the cover sleeve connector **45** is moved proximally and thus the detection pin is further moved proximally. This further movement is detected by a further TMR position sensor which triggers the controller in the electronic module to start the injection. That means the controller drives the electric motor **91** in dispensing direction and thus rotates the threaded rod **41** and thereby moves the plunger rod **40** in distal direction. As the trigger sleeve spring **53** biases the trigger sleeve connector **45** on the proximal shoulder of the plunger rod **40** (alternatively the trigger sleeve connector **45** is immovably connected to the plunger rod **40**) the trigger sleeve **50** is also moved in distal dispensing direction. The distal flange **49** of the plunger rod **40** moves the piston **16** inside the syringe **15** in distal direction and thus dispenses liquid drug through the injection needle out of the syringe **15**.

[0155] The distal movement of the plunger rod **40** moves the trigger sleeve via trigger sleeve connector **51** in distal direction to its initial position. That means the trigger arms **54** can move radially outwards back into their non-deflected position. However, as the cover sleeve **20** is in a retracted proximal position the locking arms **32** are prevented from being moved back into the locking position.

[0156] The controller displays a holding time indicating a time period during that the user has to hold the autoinjector **1** onto the injection site after the injection to ensure that the drug is completely administered and absorbed. The controller may display a down counter informing the user about the remaining holding time. Once the holding time is elapsed a notification is shown to the user that the autoinjector can be removed from the injection site. Alternatively, the end of the

holding time may be indicated only by a LED.

[0157] When the user removes the autoinjector **1** from the injection site compressed cover sleeve spring **46** can release and thus it can move the cover sleeve **20** back distally into the needle covering position.

[0158] During dispensing the trigger sleeve **50** was moved in its initial distal and unlocking position in which the trigger arms **54** are not deflected anymore and thus do not contact the locking arms **32** of the syringe holder **30**. That means the locking arms **32** are no longer completely pressed radially inwards and upon movement of the cover sleeve **20** into its covering position the locking arms **32** are able to switch back into their locking position in which the free end of the arms **32** abuts the contact surface inside the cover sleeve **20** thereby locking the cover sleeve **20** in its covering position as in the state shown in FIG. 4.

[0159] Furthermore, as the trigger arms **54** are not deflected anymore and thus do not contact the distal stop surface of the syringe holder **30** the entire syringe unit **2** can be pulled out of the drive unit housing **80** by the user. The user can now discard the syringe unit **2** and the drive unit **3** is ready to be loaded with a new syringe unit.

Second embodiment

[0160] In the following a second embodiment of the present disclosure is described with respect to FIGS. **6a** to **13**. The drug delivery device of the second embodiment is a semi-reusable autoinjector **100** including a drive unit **103** and a syringe unit **102**. The differences to the first embodiment autoinjector will be described in detail.

[0161] The syringe unit **102** of the second embodiment is shown in FIGS. **6a** and **6b** is similar to the syringe unit of the first embodiment. FIG. **6a** depicts a perspective view of the syringe unit **102** with the cap **110** and FIG. **6b** depicts the syringe unit **102** without cap.

[0162] In contrast to the first embodiment the syringe holder **130** of the syringe unit **102** of the second embodiment includes two cap holding tongues **135** extending axially and deflectable radially inwards. The two oppositely arranged tongues **135** have each a free end including a radially outwards extending protrusion **136** adapted to interact with a cap actuation nose **179** as described below. Each tongues **135** further has two ledges **137** next to protrusion **136** on both sides. The ledges **137** engage corresponding recesses in the cap **110** to hold the cap **110** on the cover sleeve **120** by a releasable snap-fit connection.

[0163] FIG. 7 depicts a sectional view of a drive unit **103** of the second embodiment. In contrast to the first embodiment described above the drive unit **103** includes an intermediate sleeve **160** radially positioned between a distal housing insert **181** and a trigger sleeve **150**.

[0164] FIG. 7 depicts a perspective view of the intermediate sleeve **160**. As shown in FIG. 7 the intermediate sleeve **160** includes two axially extending levers **170** that are connected to a sleeve-shaped element similar to the above described gripper sleeve of the first embodiment. The intermediate sleeve **160** further includes two clamping arms **161** (see FIG. 7) for holding the syringe unit **102** as described with respect to the first embodiment. Each clamping arm **161** is connected to a spring **162**. The spring **162** is supported on a sleeve mechanism **183** and biased the clamping arms **161** in a holding position. The intermediate sleeve **160** is axially fixed relative to the housing **180** and to the distal housing insert **181**.

[0165] Furthermore, as shown in FIGS. 7 and 8 the intermediate sleeve **160** includes on its levers **170** two radially inwards protruding actuation noses **178** adapted to abut locking arms **132** of an inserted syringe unit **102**. Additionally, on a distal end each lever **170** includes a radially inwards protruding cap actuation nose **179** which are adapted to interact with the deflectable cap holding tongues **135** of the syringe unit **102**. The trigger sleeve **150** includes lateral openings allowing the actuation nose **178** and the cap actuation nose **179** to extend through to reach the locking arms **132** and the tongues **135**, respectively. The levers **170** include radially outside the actuation nose **178** counter cams **171** extending in a plane perpendicular to the longitudinal axis and including contact surfaces on a distal and on a proximal side as will be described in detail below. Correspondingly,

the levers **170** include radially outside the cap actuation noses **179** counter cap cams **175** with contact surfaces. The contact surfaces of the counter cam **171** and of the counter cap cams **175** are adapted to interact with contact surfaces of control cams **151** and cap cams **155** of the trigger sleeve **150**.

[0166] FIG. **9a** shows the autoinjector **100** in an unlocked state in a sectional view wherein the cut runs along the longitudinal axis. The syringe unit **102** is inserted into the drive unit **103** but not yet locked to the drive unit **103**. FIG. **9b** shows the same state but the cut for the sectional view is offset parallel to the cut of FIG. **9a** to show the control cams **151**, cap cams **155** and counter cams **171** and counter cap cams **175**.

[0167] As shown in FIG. **9b** the trigger sleeve **150** control cams **151** are arranged on an outer portion oppositely to each other and extending in a plane perpendicular to the longitudinal axis. The control cams **151** are adapted to interact with the above mentioned counter cams **171** of the intermediate sleeve **160**. Additionally, the cap cams **155** of trigger sleeve **150** are arranged on a distal end portion oppositely to each other and are adapted to interact with the corresponding counter cap cams **175** of the intermediate sleeve **160**.

[0168] Further as shown in FIG. **9b** the control cams **151** include on a distal side a first sloped contact surface **152** adapted to engage a second sloped contact surface **172** on a proximal side of the counter cam **171** of the intermediate sleeve **160**. Furthermore, the control cam **151** includes on a proximal side a third sloped contact surface **153** adapted to engage a fourth sloped contact surface **173** on a distal side of the counter cam **171**. As shown in FIG. **9b** all sloped contact surfaces **152**, **153**, **172**, **173** have the same angle to the longitudinal axis.

[0169] Similarly, the cap cams **155** include a distal surface **156** and a proximal surface **157** and the counter cap cams **175** of the intermediate sleeve **160** include a counter distal surface **176** and a counter proximal surface **177**. As it can be seen in FIG. **9b** the distal and proximal surfaces of the cams **155**, **175** are oppositely angled such that the cap cams **155** and counter cap cams **175** are shaped as a truncated pyramid in a plane parallel to the longitudinal axis.

[0170] The function of the second embodiment autoinjector **100** will be described in the following.

[0171] In a first step, the user inserts a proximal portion of the syringe unit **102** into the housing of the drive unit **103**. Upon contact with the syringe unit ledges **133** the clamping arms **161** deflect radially and snap onto the proximal ledges **133** of the syringe holder **130** as described in the first embodiment. The syringe connector **147** biased by the syringe connector spring **148** abuts a proximal rim portion or flange of the syringe **15** and thus ensures that the syringe is pressed against a bearing surface inside the syringe holder **130**.

[0172] The position sensor detects that the syringe unit **102** has been inserted and provides a corresponding signal to the controller. Subsequently, after a successful medication verification as described above with respect to the first embodiment the controller drives the electric motor **191** to automatically retract the trigger sleeve **150** via trigger sleeve connector **158** in a proximal direction.

[0173] That means the trigger sleeve **150** is axially moved relative to the intermediate sleeve **160** which does not move. Therefore, the control cams **151** and the cap cams **155** of the trigger sleeve **150** move from an initial distal position as shown in FIG. **9b** to a proximal position as shown in FIG. **10**. During the proximal movement of the trigger sleeve **150** the control cams **151** thus pass radially inwards the counter cams **171** of the intermediate member **160**. The cap cams **155**, however, come into contact with their proximal sloped surface **157** with the distal sloped surface **176** of the counter cap cam **175** during trigger sleeve **150** movement relative to the intermediate sleeve **160**.

[0174] Due to the movement of the trigger sleeve **150** the cap cam proximal sloped surface **157** slide along the counter cap cam distal surface **176** and thereby forcing the levers **170** of intermediate member **160** to deflect radially inwards which moves the cap actuation noses **179** radially inwards too. As the trigger sleeve **150** is further shifted proximally the cap cams **155** of the trigger sleeve passes above the counter cap cam **175** and thus keeps the levers **170** in a radially

inwards deflected position as shown in FIGS. **10** and **11**.

[0175] In case the medication verification is not successful (e.g., an incorrect syringe unit was inserted) the controller drives the motor to move the trigger sleeve **150** in distal direction to release the syringe unit **102** from the drive unit **103** as described below with respect to the last step after an injection.

[0176] As the cap actuation noses **179** are now moved radially inwards the noses **179** press onto the protrusion **136** of the cap holding tongues **135** and thus deflect the holding tongues **135** radially inwards. This releases the ledges **137** from the cap **110** and thus the snap-fit connection between the tongues **135** and the cap **110** is released and the user can remove the cap **110** from the cover sleeve **120**.

[0177] Subsequently, a cover sleeve lock is unlocked and the syringe unit **102** is locked to the drive unit **103**. That is, the controller drives the motor **191** in dispensing direction causing the plunger rod **140** to move in distal direction. As the trigger sleeve connector **158** is pressed onto a proximal end of the plunger rod **140** by the trigger sleeve spring **159** (or alternatively the trigger sleeve connector is fixedly connected to the plunger rod) the trigger sleeve **150** is moved via trigger sleeve connector **158** in distal direction too.

[0178] This brings the first contact surfaces **152** of the control cams **151** into contact with the second contact surfaces **172** of the counter cams **171** of the intermediate member **160**. As the first and second contact surfaces **152**, **172** are sloped the two surfaces slide along each other and as the trigger sleeve **150** is axially guided and cannot move in radial direction the control cams **151** press the counter cams **171** radially inwards such that control cams **151** pass radially outwards the counter cams **171** (FIG. **11**). This bends or deflects a middle portion of the levers **170** of the intermediate sleeve **160** radially inwards and thereby moves the actuation noses **178** radially inwards as shown in FIG. **11**. As a consequence, the actuation noses **178** press on the deflectable locking arms **132** of the syringe holder **130** thereby move the locking arms **132** radially inwards, see FIG. **11**. This in turn moves the free end of the locking arms **132** out of a locking position and the arms **132** no longer contact the stop surface of the cover sleeve **120**. The cover sleeve **120** is thus unlocked and is able to move proximally relative to the syringe holder **130** and relative to the housing **180**.

[0179] At the same time and in a manner analogous as in the first embodiment the radially inwards deflected actuation noses **178** abut a distal stop surface of the syringe holder **130** and thereby lock the syringe holder **130** relative to the trigger sleeve **150** such that a movement of the trigger sleeve **150** is transferred to the syringe unit **2**. Thus, in this state syringe unit **102** is locked inside the drive unit **3**.

[0180] During the above described distal movement of the trigger sleeve **150**, the cap cams **155** are moved relative to the counter cap cams **175** of the intermediate sleeve **150**. However, the cap cams **155** still press the counter cap cams **175** radially inwards and thus keep the cap actuation noses **179** in the radially inwards position as depicted in FIG. **11**.

[0181] The controller displays a message on the display or activates LEDs to inform the user that the autoinjector is ready for and injection.

[0182] FIG. **11** shows the above described state of the autoinjector **100** where the cap **110** is released and the cover sleeve **120** is unlocked and the syringe unit is locked by the radially inwards moved actuation noses **178**.

[0183] As described with respect to the first embodiment the injection is started by a push on skin release. FIG. **12** depicts the state where the cover sleeve **120** is moved proximally out of the covering position if the user presses the autoinjector **100** onto the skin. The cover sleeve **120** moves the cover sleeve connector **145** proximally and thereby compressing the cover sleeve spring **146**. In a proximal end position a TMR position sensor detects via detection pin the proximal position of the retracted cover sleeve connector **145** and provides a start signal to the controller. The latter starts the injection by driving the motor **191** in dispensing direction.

[0184] The motor **191** rotates the threaded rod **141** and moves the non-rotating plunger rod **140** in distal direction and thus moves the trigger sleeve **150** via trigger sleeve connector **158** in distal direction. That means the trigger sleeve **150** with its control cams **151** and cap cams **155** is moved relative to the intermediate sleeve **160**. The control cams **151** and cap cams **155** are moved from the position shown in FIG. **11** to the position shown in FIG. **12** wherein the control cams **151** and cap cams **155** pass and release the counter cams **171**, **175** of the intermediate sleeve **160** and thus allow the intermediate sleeve levers **170** to deflect back into their initial non-deflected state. Thereby the actuation noses **178** and cap actuation noses **179** are move radially back outwards into their initial state. That would in turn return the syringe holder locking arms **132** towards their locking position. However, as the cover sleeve **120** is in a proximal retracted position the locking arms **132** cannot move completely radially outwards.

[0185] The motor **191** further drives the plunger rod **140** in distal dispensing direction and the plunger rod **140** moves the piston **16** inside the syringe **15** distally to dispense the liquid drug until the drug is completely dispensed. Subsequently, the controller displays a holding time counter as described with respect to the first embodiment.

[0186] When the user removes the autoinjector **100** from the injection site the cover sleeve **120** can move back into the covering position by a force of the compressed the cover sleeve spring **146**. As the levers **170** of the intermediate sleeve **160** and thus the actuation nose **178** and the cap actuation nose **179** are in the retracted state the locking arms **132** of the syringe holder **130** can move back radially outwards once the cover sleeve **120** is moved back distally in its covering position. The cover sleeve **120** is therefore locked again by the locking arms **132** and thus is prevented from a movement in proximal direction.

[0187] The distal movement of the cover sleeve **120** moves the cover sleeve connector **145** too and the position sensor detects the movement via detection pin. Upon confirmation of the user via button the controller releases the lock of the syringe unit **102**. For this purpose, the controller rotates the motor **191** in reverse direction to move the trigger sleeve **150** in a proximal end position. The proximal movement of the trigger sleeve **150** brings a proximal (third) contact surfaces **153** of the control cams **151** into contact with distal (fourth) contact surfaces **173** of the counter cams **171**. Due to the sloped angle the counter cams **171** are moved radially outwards and thus move the levers **170** into their initial position (if the levers **170** are not already fully deflected back in the initial position). The control cams **151** and the cap cams **155** then pass the counter cams **171** and counter cap cams **175** respectively to a proximal position such that the cams of the trigger sleeve **150** do not contact the counter cams of the intermediate sleeve **160** anymore and thus levers **170** are not deflected any more.

[0188] If the trigger sleeve **150** is moved towards its proximal end position a proximal end wall **154** of the trigger sleeve **150** abuts a sloped release surface **163** of each clamping arm **161**. If the trigger sleeve **150** is further moved in its proximal end position the end wall **154** presses the clamping arms **161** radially outwards and thus compressing the arm springs **162** as shown in FIG. **13**. This releases the clamping arms from the ledges **133** and thus the syringe unit **102** is released from the drive unit **103**. That means the compressed syringe connector spring **148** can relax thereby moving by the syringe connector **147** the entire unlocked syringe unit **102** in distal direction. The user can then remove the syringe unit **102** out of the drive unit housing **180**. The controller drives the motor to move the trigger sleeve **150** distally back in its initial position. The autoinjector **100** is now ready to be loaded with a new syringe unit **102**.

[0189] FIG. **14** depicts a third embodiment with an autoinjector **200** according to the disclosure. In contrast to the second embodiment the drive unit **203** of the autoinjector **200** includes a slider **210** that can be shifted along the longitudinal axis by the user in order to move the syringe unit **202** out of the drive unit after an injection. Hence, in contrast to the above described last step (with respect to FIG. **13**) the clamping arms are not deflected by a trigger sleeve movement but by a manual movement of the syringe unit by the user. All other features of the third embodiment are identical

the above described second embodiment.

[0190] FIGS. 15 to 17 show a further embodiment of the disclosure. FIG. 15 shows a system according to the disclosure showing the injection device 301, an exchangeable cartridge unit 310 which may be inserted into the injection device 301, and smart phone 320. The NFC tag 311 on the cartridge unit housing 310a is depicted as a symbolic antenna. FIG. 16 shows the injection device 301 with the outer distal housing 301b removed, such that the near field communication module, i.e., the NFC receiver and transmitter 302 becomes visible-also depicted as symbolic antenna. The cartridge module 310 includes a prefilled syringe 312 (see FIG. 17). Within the syringe a piston 314 is disposed, which is movable between proximal position (as in FIG. 17) and a distal position where piston reaches the shoulder of the syringe body. As the piston is moved into distal direction, fluid medicament is ejected through the syringe cannula 313. Before use, the cannula is protected by a needle shield 315 which is removed by the needle shield removing unit 316 when the injection device 301 is to be used to deliver medicament.

[0191] Injection device 301 is a motor driven autoinjector. It includes an electronic controller, a drive assembly including the motor and the NFC receiver and transmitter. A person skilled in the art is aware of motor driven autoinjectors, e.g., US 20200164145 A1 is a known example. The injection device 301 includes an inner housing 301c and outer housings, namely the proximal outer housing 301a and distal outer housing. After insertion of the cartridge unit 310 is detected by the injection device, e.g., by sensors in communication with the controller, the controller activates the NFC receiver and transmitter 302 to read the information contained in the NFC Tag 311. The read information can be used by the injection device for the preparation of the medicament injection. The controller may further use the read information to check the type of medicament and e.g., the shelf life. If the expiration date is reached the injection device 301 may emit a signal to the person who intends to use the device-therefore improving safety of the medicament use.

[0192] The NFC receiving and transmitting 302 device may be used to transmit data and further information to the smart phone 320 which includes an appropriate reading unit. In order to transmit data and further information from the injection device 301 to the smart phone 320 the controller of the injection device 301 sets the NFC receiver and transmitter 302 into the so-called card emulation mode and provide the data and further information the NFC receiver and transmitter 302. The smart phone 320 and the injection device 301 are then moved to close proximity or directly adjacent which allows the reading unit of the smart phone to extract the data and further information from the NFC receiver and transmitter 302. The smart phone 320 is configured with software to analyze and process the received data and further information. For example transmitted error messages may be displayed on the display 321 of the smart phone 320. Moreover, the smart phone 320 is configured to forward at least part of the read data and further information, notably in processed form, to further devices, e.g., distant servers. This may be done through a local area network (LAN) connection, particularly through wireless LAN (WLAN) or a cellular or mobile network.

[0193] While the disclosure is described in detail in the drawings and foregoing description, such description is to be considered illustrative or exemplary and not restrictive. Variations to the disclosed embodiments will be understood and effected by those skilled in the art and practising the claimed disclosure, from a study of the drawings, the disclosure, and the appended claims.

[0194] In the claims, the word “including” does not exclude other elements or steps, and the indefinite article “a” or “an” does not exclude a plurality. For example “an arm” does not exclude the fact that there may be two arms that functionally or structurally fulfill the purpose of “an arm”. The mere fact that certain elements or steps are recited in distinct claims shall not preclude the existence of further meaningful combinations of these elements or steps.

TABLE-US-00001 LIST OF DESIGNATIONS 1 autoinjector 2 syringe unit 3 drive unit
4 button 10 cap 11 deflectable tongues 12 holder 15 prefilled syringe 16 piston 17 rigid
needle shield 18 injection needle 20 cover sleeve 21 first opening 22 second opening 23
indentation 24 stop surface 25 label 26 RFID tag 27 release surface 30 syringe holder 31

longitudinal rail 32 locking arms 33 ledge 34 bearing surface 40 plunger rod 41 threaded rod
45 cover sleeve connector 46 cover sleeve spring 47 syringe connector 48 syringe connector
spring 49 flange 50 trigger sleeve 51 trigger sleeve connector 52 ledge 53 trigger sleeve
spring 54 trigger arm 55 arm sloped contact surface 56 free end 60 gripper sleeve 61
clamping arms 62 clamp 64 gripper sleeve spring 65 counter release surface 80 housing 81
distal housing insert 82 contact surface 83 sleeve mechanism 84 wall 85 proximal housing
insert 86 protrusion 91 electric motor 92 pinion 93 gear 94 battery 100 autoinjector 102
syringe unit 103 drive unit 110 cap 120 cover sleeve 130 syringe holder 132 locking arms 133
ledges 135 cap holding tongues 136 protrusion 137 ledge 140 plunger rod 141 threaded rod 145
cover sleeve connector 146 cover sleeve spring 147 syringe connector 148 syringe connector spring
150 trigger sleeve 151 control cam 152 first contact surface 153 third contact surface 154 end wall
155 cap cam 156 cap distal surface 157 cap prox. surface 158 trigger sleeve connector 159 trigger
sleeve spring 160 intermediate sleeve 161 clamping arms 162 spring 163 release surface 170 lever
171 counter cam 172 second contact surface 173 fourth contact surface 175 counter cap cam 176
counter distal surface 177 counter prox. surface 178 actuation nose 179 cap actuation nose 180
housing 181 distal housing insert 183 sleeve mechanism 191 motor 200 autoinjector 202 syringe
unit 203 drive unit 210 slider 301 autoinjector 301a proximal outer housing 301b distal outer
housing 301c inner housing 302 NFC receiver and transmitter 303 cartridge unit cavity 310
cartridge unit 310a cartridge unit housing 311 NFC tag 312 prefilled syringe 313 syringe cannula
314 syringe piston 315 needle shield 316 needle shield removing unit 320 smart phone 321 display

Claims

1. A reusable drive unit of a drug delivery device for dispensing a liquid drug from a reservoir through a needle of a reservoir unit, wherein the drive unit is configured to be connected to the reservoir unit and comprises: a housing defining a longitudinal axis; an electronic module arranged in an interior of the housing, the electronic module comprising a controller; a connector arranged in the housing interior, the connector biased in a distal direction by a spring and configured to be axially movable; a detection element connected to the connector, wherein the connector and detection element are axially movable relative to the housing during attachment of a reservoir unit to the drive unit; and a position sensor arranged in the housing interior and communicatively coupled to the controller and configured to sense an axial movement of the detection element, wherein when the connector and the detection element are axially moved, the sensor is configured to transmit a signal to the controller corresponding to a sensed axial movement of the detection element such that the controller detects a presence of a reservoir unit in the drive unit.
2. The drive unit of claim 1, wherein the detected presence of the reservoir unit in the drive unit corresponds to the reservoir unit being in a locked state with the drive unit.
3. The drive unit of claim 2, wherein the drive unit is configured to establish the locked state with the drive unit by a releasable connection in a snap-fit, a bayonet connection or a thread.
4. The drive unit of claim 3, wherein the connector comprises a distally directed surface configured to contact a proximally directed surface of the reservoir unit to cause the axial movement of the detection element.
5. The drive unit of claim 1, wherein the connector comprises a distally directed surface configured to contact a proximally directed surface of the reservoir unit to cause the axial movement of the detection element.
6. The drive unit of claim 1, wherein the drive unit comprises a plunger rod coaxially arranged within the connector and the spring, the plunger rod configured to be moved through an opening defined in the connector in the distal direction in a dispensing movement to cause the liquid drug to be dispensed from the reservoir of the reservoir unit.
7. The drive unit of claim 6, wherein the plunger rod is non-rotatable relative to the housing during

the dispensing movement.

8. The drive unit of claim 7, wherein the plunger rod is in a threaded engagement with an automatic drive configured cause the plunger rod to be moved in the dispensing movement.

9. The drive unit of claim 1, wherein the position sensor is configured as a magnetic sensor.

10. A drug delivery device comprising a reusable drive unit and a reservoir unit, comprising: a drive unit housing defining a longitudinal axis; an electronic module arranged in an interior of the housing, the electronic module comprising a controller; a connector arranged in the housing interior, the connector biased in a distal direction by a spring and configured to be axially movable; a detection element connected to the connector, wherein the connector and detection element are axially movable relative to the housing during attachment of the reservoir unit to the drive unit; and a position sensor arranged in the housing interior and communicatively coupled to the controller and configured to sense an axial movement of the detection element, wherein the connector comprises a distally directed surface configured to contact a proximally directed surface of the reservoir unit to cause the axial movement of the detection element, and wherein when the detection element is axially moved, the sensor is configured to transmit a signal to the controller corresponding to a sensed axial movement of the detection element such that the controller detects a presence of the reservoir unit in the drive unit.

11. The drug delivery device of claim 10, wherein the detected presence of the reservoir unit in the drive unit corresponds to the reservoir unit being in a locked state with the drive unit.

12. The drug delivery device of claim 11, wherein the drive unit is configured to establish the locked state with the drive unit by a releasable connection in a snap-fit, a bayonet connection or a thread.

13. The drug delivery device of claim 12, wherein the position sensor is configured as a magnetic sensor.

14. The drug delivery device of claim 10, wherein the drive unit comprises a plunger rod coaxially arranged within the connector and the spring, the plunger rod configured to be moved through an opening defined in the connector in the distal direction in a dispensing movement to cause a liquid drug to be dispensed from a reservoir of the reservoir unit.

15. The drug delivery device of claim 14, wherein the plunger rod is non-rotatable relative to the housing during the dispensing movement, and wherein the plunger rod is in a threaded engagement with an automatic drive configured cause the plunger rod to be moved in the dispensing movement.

16. The drug delivery device of claim 10, wherein the reservoir unit comprises a needle cover sleeve with a retracted position and a needle covering position.

17. The drug delivery device of claim 16, wherein the position sensor is configured to transmit a signal to the controller corresponding to a sensed movement of the needle cover sleeve.

18. The drug delivery device of claim 17, wherein a movement of the needle cover sleeve is guided by the reservoir unit.

19. The drug delivery device of claim 10, wherein the reservoir unit comprises a tag and the drive unit comprises a tag reader for reading the tag of reservoir unit.

20. The drug delivery device of claim 19, wherein the tag is configured as an NPC tag or a RFID tag.
